

A Reproducible Behavior-Cloning-to-PPO Pipeline for Vision-Based UAV Navigation in Randomized Cities

[Author Name(s)]

[Department, Institution, City, Country; e-mail: corresponding@author.edu]

Abstract—This paper presents a reproducible behavior-cloning-to-proximal-policy-optimization (BC-to-PPO) pipeline for fixed-height unmanned aerial vehicle (UAV) navigation in randomized high-rise scenes rendered by NVIDIA Isaac Sim. An A* expert accepts only reachable layouts and generated 36 successful, synchronized episodes containing 2,649 image-action samples. A frozen convolutional autoencoder compresses clean first-person RGB images to 64 dimensions, which are fused with an eight-dimensional goal and motion state. BC attained a held-out action mean-squared error of 0.01777 and 45% closed-loop success on 20 comparison seeds. Exact-weight BC initialization followed by 100,000 PPO steps reached 54% success over 100 seeded episodes. The results establish an auditable baseline while exposing collision and visual-domain-shift limitations.

I. INTRODUCTION

Learning vision-based navigation from simulator demonstrations is attractive because it reduces manual labeling, but temporal misalignment, privileged visual markers, and incompatible policy interfaces can make reported performance difficult to reproduce. We address these engineering gaps with a deterministic environment, an A*-verified expert [1], a same-step observation-action contract, and an actor whose weights transfer exactly from BC to PPO [2], [3]. The scope is deliberately limited to planar body velocity and yaw-rate control at fixed altitude; rigid-body dynamics and hardware flight are not claimed.

II. METHODS AND RESULTS

For each external seed, the environment samples a city and resamples deterministically until A* confirms a collision-free start-to-goal route. Debug geometry is excluded from the policy camera. At time t , the collector captures a 128×72 RGB observation, computes and stores the expert action, and only then advances the simulation. Episode-level splitting produced 24/6/6 training/validation/test episodes with 1,768/447/434 samples, preventing frame leakage.

The frozen encoder maps normalized RGB to a 64-D latent vector. The remaining eight inputs are body velocity, body-frame goal direction, normalized goal distance, and the previous action. A 72–128–128–3 multilayer perceptron outputs normalized forward velocity, right velocity, and yaw rate; their physical limits are 1.0 m/s, 0.8 m/s, and 1.0 rad/s. PPO preserves this actor interface and uses clipped updates with dense goal-progress reward, small step and smoothness costs, and symmetric terminal success/collision rewards.

TABLE I. CLOSED-LOOP SEEDED-CITY EVALUATION

Policy	Updates	n	Success	Coll./TO
BC	0	20	45%	55/0%
PPO	8,192	20	40%	60/0%
PPO	50,000	50	52%	44/4%
PPO	100,000	100	54%	43/3%

All seed sequences start at 50,000. Coll./TO denotes collision/timeout. Different n values preclude a controlled significance claim.

All expert episodes were successful and marked synchronized. BC checkpoint reload and exact PPO actor initialization were verified, and PPO changed the actor weights. As Table I shows, the longer runs improved over the 8,192-step PPO run, but success plateaued near one half and collision remained the dominant failure. Moreover, the encoder was trained on an earlier RGB collection, whereas demonstrations used marker-free images; this mismatch is a plausible limitation rather than evidence of robust navigation.

III. CONCLUSION

The proposed pipeline turns reachable scene generation, synchronized expert collection, BC, and PPO refinement into one reproducible UAV navigation baseline. The recorded artifacts support integration correctness and modest closed-loop performance, not deployment readiness. Future work will retrain the encoder on the marker-free distribution, gate checkpoints on a common evaluation set, add DAGger-style recovery data, and validate learned control with rigid-body dynamics before PX4 or hardware experiments.

REFERENCES

- [1] P. E. Hart, N. J. Nilsson, and B. Raphael, “A formal basis for the heuristic determination of minimum cost paths,” *IEEE Trans. Syst. Sci. Cybern.*, vol. 4, no. 2, pp. 100–107, 1968.
- [2] D. A. Pomerleau, “ALVINN: An autonomous land vehicle in a neural network,” in *Proc. Adv. Neural Inf. Process. Syst.*, 1989, pp. 305–313.
- [3] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, “Proximal policy optimization algorithms,” *arXiv:1707.06347*, 2017.